

SUWALKI, Poland

WMO: 121950

Lat: 54.1308N

Lon: 22.9489E

Elev: 185

StdP: 99.12

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-19.1	-15.9	-21.4	0.6	-18.9	-18.2	0.8	-15.5	12.5	2.9	10.8	1.8	1.8	40	0.586

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.2	28.7	19.8	26.7	18.8	25.1	17.9	21.0	26.4	19.9	25.2	18.9	23.3	3.3	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.1	14.2	23.6	18.1	13.3	22.2	17.2	12.6	21.1	61.5	26.2	57.8	25.3	54.5	23.3	25.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
9.3	8.2	7.2	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
			-22.6	31.6	3.9	1.7	-25.4	32.8	-27.7	33.8	-30.0	34.8	-32.8	36.0
			-22.8	22.8	3.9	1.2	-25.6	23.6	-27.9	24.3	-30.1	25.0	-32.9	25.9
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	7.1	-3.9	-3.0	0.8	7.2	12.2	15.5	18.0	17.3	12.5	7.1	2.4	-1.8
	DBStd	8.84	5.87	5.62	4.52	4.32	4.07	3.36	3.11	3.05	3.28	3.85	4.17	5.41
	HDD10.0	1925	432	363	287	105	23	1	0	0	13	105	229	367
	HDD18.3	4186	690	597	544	334	193	98	45	57	175	348	479	625
	CDD10.0	857	0	0	1	21	92	165	249	225	89	16	0	0
	CDD18.3	78	0	0	0	0	4	13	36	24	1	0	0	0
	CDH23.3	796	0	0	0	9	53	142	325	247	21	0	0	0
	CDH26.7	173	0	0	0	1	4	26	84	56	2	0	0	0
Wind	WSAvg	3.5	3.9	3.9	3.9	3.5	3.3	3.2	3.0	2.8	2.9	3.5	3.8	3.9
Precipitation	PrecAvg	594	32	26	33	37	54	73	80	68	51	49	50	41
	PrecMax	829	103	54	78	94	105	149	176	181	125	192	121	78
	PrecMin	394	7	1	6	5	14	22	19	10	12	3	23	7
	PrecStd	94	18	13	17	19	23	30	42	41	25	36	22	17
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.4	7.9	15.0	24.4	27.2	29.6	31.5	31.1	26.0	19.1	11.9	8.7
		MCWB	6.5	6.3	9.1	15.4	18.0	20.4	20.8	20.9	17.5	14.2	10.3	7.7
	2%	DB	5.3	6.4	11.7	20.7	24.8	26.9	29.1	28.1	22.8	16.4	10.1	7.0
		MCWB	4.6	5.0	7.3	13.0	16.9	18.7	20.1	19.3	16.6	13.2	9.0	6.1
	5%	DB	3.9	4.7	9.1	17.8	22.6	24.8	26.8	26.1	20.5	14.3	8.7	5.6
		MCWB	3.2	3.7	6.0	10.9	15.6	17.6	19.6	18.4	15.5	12.1	7.8	4.9
	10%	DB	2.5	3.4	6.8	15.3	20.5	22.8	24.8	24.1	18.6	12.8	7.7	4.2
		MCWB	2.0	2.6	4.8	9.8	14.4	16.6	18.4	17.6	14.4	11.2	7.0	3.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.7	6.6	9.9	15.9	19.1	21.6	22.7	22.0	19.0	15.2	10.8	8.0
		MCDB	7.3	7.5	13.4	23.6	25.3	27.5	28.6	28.0	23.0	17.9	11.7	8.5
	2%	WB	4.7	5.2	7.9	13.6	17.7	19.9	21.2	20.6	17.3	13.7	9.2	6.4
		MCDB	5.2	6.1	10.8	19.1	22.8	25.2	26.4	26.0	21.4	15.5	9.9	6.9
	5%	WB	3.2	3.9	6.3	11.9	16.4	18.5	20.2	19.5	16.0	12.5	8.0	4.9
		MCDB	3.7	4.7	8.5	16.6	21.1	23.0	25.4	24.3	19.4	14.0	8.6	5.6
	10%	WB	2.1	2.6	5.0	10.4	15.1	17.3	19.2	18.4	14.9	11.3	7.0	3.6
		MCDB	2.5	3.3	6.7	14.3	19.5	21.5	23.6	22.6	18.0	12.7	7.6	4.1
Mean Daily Temperature Range	5% DB	MDBR	4.4	5.2	7.2	10.2	11.1	10.5	10.2	10.7	9.3	6.9	4.1	4.1
		MCDBR	4.1	5.2	10.8	14.8	14.1	13.7	13.4	13.8	12.6	9.5	5.2	4.1
		MCWBR	3.8	4.3	7.1	8.0	7.4	6.8	6.0	6.4	7.0	6.4	4.4	3.8
	5% WB	MCDBR	3.9	4.9	9.4	13.3	12.6	11.9	11.7	11.7	11.1	8.3	4.8	4.1
		MCWBR	3.7	4.3	6.7	7.8	7.1	6.7	5.8	5.9	6.6	6.0	4.3	3.9
Clear-Sky Solar Irradiance	taub	0.298	0.329	0.365	0.408	0.397	0.390	0.426	0.420	0.380	0.356	0.339	0.297	
	taud	2.207	2.194	2.218	2.193	2.288	2.333	2.250	2.264	2.347	2.399	2.291	2.244	
	Ebn at Noon	639	744	801	815	849	858	816	796	783	717	587	562	
	Edh at Noon	63	89	109	127	122	118	126	118	96	73	59	50	
All-Sky Solar Radiation	RadAvg	0.61	1.35	2.60	4.08	5.30	5.53	5.14	4.36	3.07	1.55	0.58	0.39	
	RadStd	0.11	0.24	0.39	0.53	0.54	0.46	0.53	0.42	0.39	0.29	0.06	0.06	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (14 neighbors)		+0.43	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+65	N/A

Nomenclature: See separate page